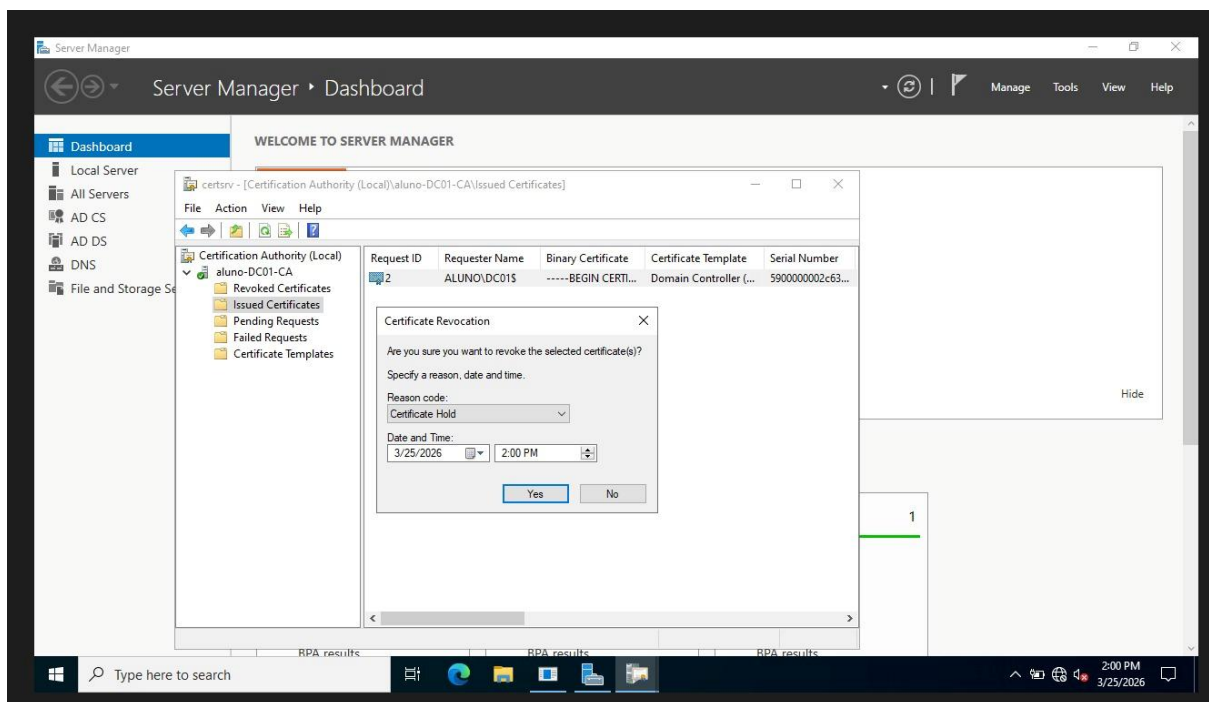
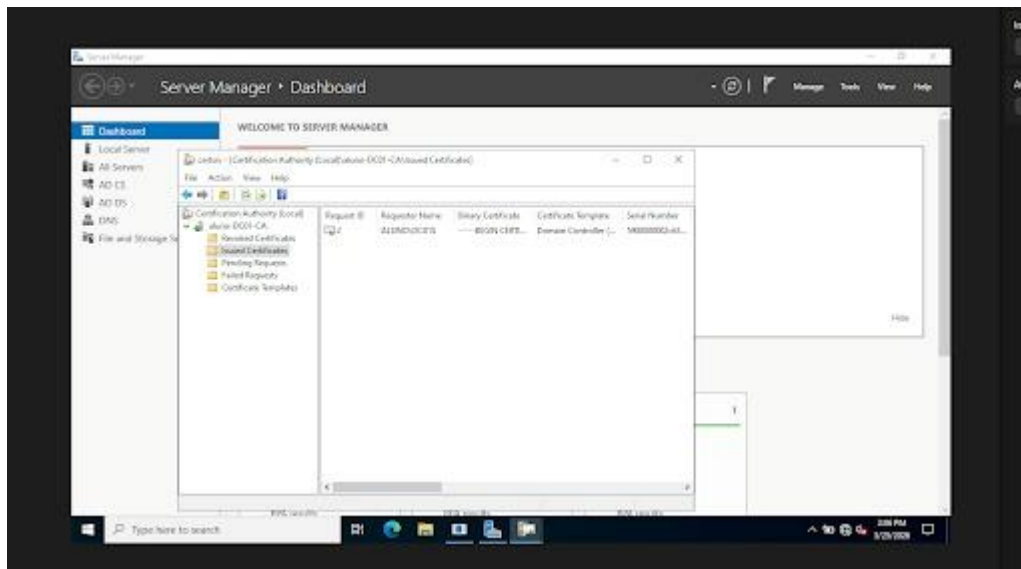


Atividade 18

Print da atividade 9.6: Importação de chaves públicas de terceiros para o chaveiro local do GPG



Print da atividade 9.7: Criptografia e descriptografia de arquivos utilizando o algoritmo simétrico para proteção de dados



Print da atividade 9.8: Uso de criptografia assimétrica para garantir que apenas o destinatário específico leia o arquivo

```
root@cleydson: ~
Sessão Ações Editar Exibir Ajuda
(cleydson@cleydson)-[~]
$ xrandr --output HDMI-1-0 --mode 1920x1080 --rotate right --right-of eDP-1
(cleydson@cleydson)-[~]
$ sudo -i
# Senha: mpsas
[sudo] senha para cleydson:
(cleydson@cleydson)-[~]
$ openssl s_client -connect www.google.com:443 -status
Connecting to 2001:4860:482d:7700::
CONNECTED(00000003)
depth=2 C=US, O=Google Trust Services LLC, CN=GTS Root R1
verify return:1
depth=1 C=US, O=Google Trust Services, CN=WR2
verify return:1
depth=0 CN=www.google.com
verify return:1
OCSP response: no response sent
-----
Certificate chain
 0 s:CN=www.google.com
  i:C=US, O=Google Trust Services, CN=WR2
  a:PKEY: EC, (prime256v1); sigalg: sha256WithRSAEncryption
  v:NotBefore: Feb 23 18:21:55 2026 GMT; NotAfter: May 18 18:21:54 2026 GMT
 1 s:C=US, O=Google Trust Services, CN=WR2
  i:C=US, O=Google Trust Services LLC, CN=GTS Root R1
  a:PKEY: RSA, 2048 (bit); sigalg: sha256WithRSAEncryption
  v:NotBefore: Dec 13 09:00:00 2023 GMT; NotAfter: Feb 20 14:00:00 2029 GMT
 2 s:C=US, O=Google Trust Services LLC, CN=GTS Root R1
  i:C=BE, O=GlobalSign nv-sa, OU=Root CA, CN=GlobalSign Root CA
  a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
  v:NotBefore: Jun 19 00:00:42 2020 GMT; NotAfter: Jan 28 00:00:42 2028 GMT
-----
Server certificate
-----BEGIN CERTIFICATE-----
MIIEVzCCAz+gAwIBAgIQZ9uCNXO5ZJkZJXTQ8GRCPjANBgkqhkiG9w0BAQsFADA7
MQswCgYDVQQGEwJVUzEeMBwGA1UEChMVVR29vZ2x1IFRydXN0IFNlcnZpY2VzMQww
CgYDVQQDEwNXUjIwHhcNMjYyMTU1WWhcNMjYyNTE4MTgyMTU0WjAQMRCw
FQYDVQQDEw53d3cuZ29vZ2x1LmNvbTBZMBMGByqGSM49AgEGCCqGSM49AwEHA0IA
BAHF0FHpIIfZ/qbAZHqnPBkDMWxPs+uIH+Y8TS0XJ6zLBmsonhYiUCzyTPykLT+
TUZBZf7LxqCxyPEmKwXtUajggJCMIICPjA0BgNVHQ8BAf8EBAMCB4AwEwYDVR0l
BAwwCgYIKwYBBQUHAWwEwDAYDVR0TAQH/BAIwADAdBgNVHQ4EFgQU86fHyHbz1mz
R+nTBWZU2Iv1d0wHwYDVR0jBBgwFoAU3hse7XkV1D43JMMhu+w0OW1CsJAwWAYI
KwYBBQUHAEETDBKCEGCGCsGAQUFBzABhhVodHRwOi8vby5wa2kuZ29vZy93cJiIw
JQYIKwYBBQUHMAKGWWh0dHA6Ly9pLnBraS5nb29nLn3dyMi5jcnQwGQYDVR0RBBIw
EIT0d3d3Lmdvb2dsZS53b29wEwYDVR0gBAwwCjAIBgZngQwBAgEwNgYDVR0gFBC8w
LTARoCmgJ4YlaHR0cDovL2MucGtpLmdvb2c3IyL29RNm55cjhGMG0wLmNybDCC
AQUGCisGAQQB1nkCBAIEGfYEGfMA8QB2AJaXZL9VWJet900HaDcIQnfp8DrV9qTz
Nm5GPd8PyqnGAAABnIvzfVYAAQDAECwRQIhAISAN0vaZa/XXugboLgIF4IrMgqo
nagU1BStuonRzLrQAiAXYuxlHKVeeotKNKUI7zfVKNQdJW1pfN0m48Z9tDCGwB3
AGQRxGykEuyniRyiAi4AvKtPKAFUHjUnq+r+1QPJfc3wAAABnIvzfUYAAQDAEGw
RgThAMPek2LUGjWqKIZ3qsMI8pG58kZmXkfchIMEDnmilBV8AiEAhtvc3lSjh2mJ
Zi+XQ1VZRSPtU81ADK9+00CEmT9hJWQwDQYJKoZIhvcNAQELBQADggEBAJzXfaT1
rNSFP882m10Ljn501kHbWupvZX+KvqDbn0utdPzS6QGVt+tf0PpSKTEcahDH82kl
tN3J7H5yVvlpr8+BhZzLMPc42ckGAvmOpUYWIDTDasXMMRvBCzImeaLME16+lhloZ
BX40hz0gJLEh5S0XJGSsjzPnLC/DTBf5XD2qaDsJ+dbNBf0K0v5fLFUvKdpv737+
ITd9yUGDqw5oN+bSfBGor8seV4KdtGEdofVRX4LD0oQ57GD6kwu94F7HbHe9Dgug
XdlS/SMj62WzYBVbLtpjGM9UwBMR7Qe0SmVUECK04khs6S1AnW0X2FTFLqskLqs3
RshiQv7pN/NYHYA=
-----END CERTIFICATE-----
subject=CN=www.google.com
issuer=C=US, O=Google Trust Services, CN=WR2
-----
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: ecdsa_secp256r1_sha256
Negotiated TLS1.3 group: X25519MLKEM768
-----
SSL handshake has read 5195 bytes and written 1772 bytes
Verification: OK
-----
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Protocol: TLSv1.3
Server public key is 256 bit
This TLS version forbids renegotiation.
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)

```

Print da atividade 9.9: Criação de assinaturas digitais para comprovar a autenticidade e o não-repúdio de um documento

```
cleydson@Cleydson: ~  
Sessão Ações Editar Exibir Ajuda  
Obter:18 http://elmirror.cl/kali kali-rolling/main amd64 python3-pyqt6.qtquick amd64 6.10.2-2 [265 kB]  
Obter:15 http://kali.download/kali kali-rolling/main amd64 python3-pyqt6 amd64 6.10.2-2 [2.606 kB]  
Obter:16 http://kali.download/kali kali-rolling/main amd64 python3-pyqt6.qtqml amd64 6.10.2-2 [249 kB]  
Obter:17 http://http.kali.org/kali kali-rolling/main amd64 libqt6quickwidgets6 amd64 6.9.2+dfsg-5 [50,0 kB]  
Obter:19 http://http.kali.org/kali kali-rolling/main amd64 electrum all 4.7.0+dfsg2-1 [103 kB]  
Obter:20 http://http.kali.org/kali kali-rolling/main amd64 python3-zbar amd64 0.23.93-9+b1 [34,1 kB]  
Obtidos 10,5 MB em 4s (2.382 kB/s)  
Selecione pacote previamente não selecionado python3-python-socks.  
(Lendo banco de dados ... 472200 arquivos e diretórios atualmente instalados).  
Preparando para desempacotar .../00-python3-python-socks_2.7.2-1_all.deb ...  
Desempacotando python3-python-socks (2.7.2-1) ...  
Selecione pacote previamente não selecionado python3-aihttp-socks.  
cleydson@Cleydson: ~  
São Ações Editar Exibir Ajuda  
cleydson@Cleydson)-[~]  
electrum listaddresses  
"bc1qc6lw88vd2zyvs67kux74c46jnt2py4skek33t",  
"bc1qrht8nhrqayk0phrz6hqlzvd44kguj2l53eyhp",  
"bc1q5wm8wy4jk4pna9p3lnq3tkydj0lj2hppjv5y4",  
"bc1qkwts8u2vzt55x6rpj53nngt267dlj5m4eurtt",  
"bc1q6qzysadk5akp7rav0p73xuerlgacyy84n5mqgr",  
"bc1qer47h29p5au2h9qdm18wcmj388f3jrpp67f6rr",  
"bc1qhu7zjdj9wxxcg9fs4vcpepm8ysc09ex9m96xhp",  
"bc1qgtw4qhfp9yfy9kzsj2pagqz8s03g4je4sy0y7r",  
"bc1qfx6x8hhqwg0atmx9lmyvdh40d7jf8306t7ygag",  
"bc1qp0jzdy24jdy5rsvtmxftr2hyqr7p47qqmxhaf",  
"bc1q8h25wqq6qyl6h5548xny8fktrkm27emrzc6sl",  
"bc1qfs42a3je58sz8pjudz2ss4t3m47hp6tu3302kl",  
"bc1qlm3eluxl92uw9pa40haxg73wt6fjlxz23y970",  
"bc1qny5prhzhwqzs7n9ysczghj407jley8p434a20",  
"bc1qx394t9s32t3j730znejp8gwm1frwx9c07tzgv",  
"bc1qywyjwaa6wajdvfyfhgzv2yfyjrdmvmnk8r2tzqax",  
"bc1qvmxfenrnywn9c7qduq86y973yuuq1wc7nw3z80",  
"bc1q45gea9vkuyj5vc4lgn5p7fyvvd6jzwn2j90z4",  
"bc1qe7wlywvf5v7tr94hezsmhg04dy2645phff2kr",  
"bc1qgdnft57yza0y3d00lqm4q3rydy657fnmac9yf",  
"bc1qp2tfang4s2hnev0s1r6atzzv8a8izcnqjz8e",  
"bc1qs5wdtw",  
"bc1qfal9d4",  
"bc1qf55zfp",  
"bc1qqy4cfj",  
"bc1qyv0s7h",  
"bc1q930qhqi",  
"bc1qkee07vl",  
"bc1q87jxtri",  
"bc1q9tze645z14rkwvz2uqcv3zaw8stouznqvanzo"  
Wallet Information  
Close  
Electrum 4.7.0 - default_wallet [standard]  
File Wallet View Tools Help  
History Send Receive Channels  
Description  
Requested amount mBTC  
Expiry 1 day  
Clear Onchain Lightning  
Balance: 0. mBTC  
Configurando python3-pyqt6.qtqml (6.10.2-2) ...  
Configurando python3-zbar:amd64 (0.23.93-9+b1) ...  
Configurando python3-protobuf (3.21.12-15) ...  
Configurando libsecp256k1-6:amd64 (0.7.1-2) ...  
Configurando python3-pyqt6.qtquick (6.10.2-2) ...  
Configurando python3-qtsass (0.4.0-1) ...  
Configurando python3-electrum-ecc (0.0.6+ds-4) ...  
Configurando python3-electrum-aionostr (0.1.0-1) ...  
Configurando python3-qdarkstyle (3.2.3+ds1-3) ...  
Configurando python3-electrum (4.7.0+dfsg2-1) ...  
Configurando electrum (4.7.0+dfsg2-1) ...  
Processando gatilhos para desktop-file-utils (0.28-1) ...  
Processando gatilhos para hicolor-icon-theme (0.18-2) ...  
Processando gatilhos para libc-bin (2.42-5) ...  
Processando gatilhos para man-db (2.13.1-1) ...  
Processando gatilhos para kali-menu (2025.4.3) ...  
(cleydson@Cleydson)-[~]  
$ electrum  
(cleydson@Cleydson)-[~]  
$ electrum  
$
```

Print da atividade 9.10: Verificação de assinaturas digitais para validar que o arquivo não foi modificado no trajeto



```
cleydson@Cleydson: ~  
Sessão  Ações  Editar  Exibir  Ajuda  
(cleydson@Cleydson)~  
$ rm ~/.electrum/wallets/default_wallet  
(cleydson@Cleydson)~  
$ electrum restore "exact measure lounge throw pluck summer coin bird material olive okay kick"  
Daemon not running; try 'electrum daemon -d'  
To run this command without a daemon, use --offline  
(cleydson@Cleydson)~  
$ electrum restore "exact measure lounge throw pluck summer coin bird material olive okay kick" --offline  
{  
  "msg": "This wallet was restored offline. It may contain more addresses than displayed. Start a daemon and use load_wallet to sync its history.",  
  "path": "/home/cleydson/.electrum/wallets/default_wallet"  
}  
(cleydson@Cleydson)~  
$ electrum getunusedaddress  
Daemon not running; try 'electrum daemon -d'  
To run this command without a daemon, use --offline  
(cleydson@Cleydson)~  
$ electrum getunusedaddress --offline  
bc1qc6lw88vd2zvvs67kux74c46jtn2py4skek33t  
(cleydson@Cleydson)~  
$
```